

Chapter 6: Special Probability Densities

STAT2920: Introduction to Probability

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- ① Uniform distribution
- ② The gamma, Exponential, and Chi-Square Distributions
 - Gamma distribution
 - Exponential distribution
 - Chi-Square Distribution
- ③ Beta distribution
- ④ The normal distribution

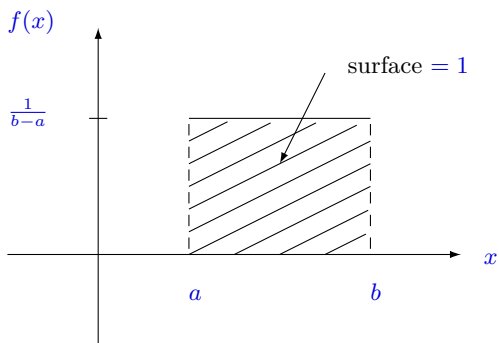
Definition 1 (Uniform Random Variable).

A random variable X is said to be a uniform random variable over the interval (a, b) if its probability density function is

$$f(x) = \begin{cases} \frac{1}{b-a} & ; \quad a < x < b \\ 0 & ; \quad \text{otherwise} \end{cases}$$

It is denoted as

$$X \sim \text{Unif}(a, b)$$



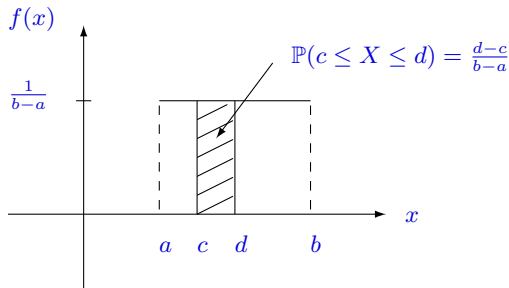
Theorem 1.

The cumulative distribution function of a uniform random variable X is

$$F_X(x) = \begin{cases} 0 & ; \quad x < a \\ \frac{x-a}{b-a} & ; \quad a \leq x < b \\ 1 & ; \quad x \geq b \end{cases}$$

If $X \sim \text{Unif}(a, b)$, then:

$$\mathbb{P}(c \leq X \leq d) = \frac{d - c}{b - a} \quad \forall \ c, d \in [a, b].$$



Proposition 1 (Properties).

For a Uniform random variable $X \sim \text{Unif}(a, b)$.

- *Moment generating function:*

$$M_X(t) = \begin{cases} \frac{e^{tb} - e^{ta}}{t(b-a)} & t \neq 0 \\ 1 & t = 0 \end{cases}$$

- *Expectation and variance are:*

$$\mathbb{E}[X] = \frac{(a+b)}{2} \quad \text{and} \quad \mathbb{V}\text{ar}(X) = \frac{(b-a)^2}{12}.$$

Proof: See next slide

- First, consider $t \neq 0$

$$\begin{aligned} M_X(t) &= \mathbb{E} \left[e^{tX} \right] = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx = \int_a^b \frac{e^{tx}}{b-a} dx \\ &= \left[\frac{e^{tx}}{t(b-a)} \right]_a^b = \frac{e^{tb} - e^{ta}}{t(b-a)} \end{aligned}$$

In the case $t = 0$, we have $\mathbb{E} [e^{0 \cdot X}] = \mathbb{E}[1] = 1$.

- Since the mean is

$$\mathbb{E}[X] = \frac{1}{b-a} \int_a^b x dx = \frac{a+b}{2},$$

and

$$\mathbb{E} [X^2] = \frac{1}{b-a} \int_a^b x^2 dx = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

it follows that

$$\text{Var} X = \frac{a^2 + ab + b^2}{3} - \frac{a^2 + 2ab + b^2}{4} = \frac{a^2 - 2ab + b^2}{12} = \frac{(b-a)^2}{12}$$

Example 2.

If X is uniformly distributed over $(0, 10)$, calculate the probability that

1. $X < 3$,
2. $X > 6$,
3. $3 < X < 8$

Solution

1.

$$\mathbb{P}(X < 3) = \int_0^3 \frac{1}{10} dx = \frac{3}{10}$$

2.

$$\mathbb{P}(X > 6) = \int_6^{10} \frac{1}{10} dx = \frac{4}{10}$$

3.

$$\mathbb{P}(3 < X < 8) = \int_3^8 \frac{1}{10} dx = \frac{1}{10}$$

Example 3.

Buses arrive at a specified stop at 15-minute intervals starting at 7 A.M. That is, they arrive at 7, 7 : 15, 7 : 30, 7 : 45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7 : 30, find the probability that he waits

- (a) less than 5 minutes for a bus;
- (b) more than 10 minutes for a bus.

Solution Let X denote the number of minutes past 7 that the passenger arrives at the stop. Since $X \sim \text{Unif}(0, 30)$, it follows that the passenger will have to wait less than 5 minutes if (and only if) he arrives between 7 : 10 and 7 : 15 or between 7 : 25 and 7 : 30. Hence, the desired probability for part (a) is

(a)

$$\mathbb{P}(10 < X < 15) + \mathbb{P}(25 < X < 30) = \int_{10}^{15} \frac{1}{30} dx + \int_{25}^{30} \frac{1}{30} dx = \frac{1}{3}$$

(b) Similarly, he would have to wait more than 10 minutes if he arrives between 7 and 7 : 05 or between 7 : 15 and 7 : 20, so the probability for part (b) is

$$\mathbb{P}(0 < X < 5) + \mathbb{P}(15 < X < 20) = \frac{1}{3}$$

Definition 1.

Let $\alpha > 0$, and consider

$$\int_0^{\infty} t^{\alpha-1} e^{-t} dt.$$

One can verify that this integral converges if and only if $\alpha > 0$. This integral is known as the Gamma function and denoted by

$$\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} e^{-t} dt.$$

Proposition 2 (Properties of Γ function).

The Gamma function satisfies the following properties:

1. $\Gamma(1) = 1$;
2. $\Gamma(2) = 1$;
3. $\Gamma(n) = (n-1)!$, for $n \in \mathbb{N}$;
4. $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$; $\forall \alpha > 0$;
5. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Proof.

1. The fact that $\Gamma(1) = \int_0^\infty e^{-t} dt = 1$ is trivial.
4. Further, Integration by parts,

$$\begin{aligned}\Gamma(\alpha) &= \int_0^\infty t^{\alpha-1} e^{-t} dt \\&= \underbrace{\left[-t^{\alpha-1} e^{-t} \right]_{t=0}^\infty}_{=0} + \int_0^\infty (\alpha-1) t^{(\alpha-1)-1} e^{-t} dt \\&= (\alpha-1) \Gamma(\alpha-1).\end{aligned}$$



Proof: Cont'd.

5.

$$\begin{aligned}
 \Gamma(1/2) &= \int_0^\infty t^{-1/2} e^{-t} dt \stackrel{z=\sqrt{2t}}{=} \int_0^\infty \sqrt{2} \exp\left(-\frac{z^2}{2}\right) dz \\
 &= 2\sqrt{\pi} \int_0^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz \\
 &\stackrel{\text{symmetry}}{=} \sqrt{\pi} \underbrace{\int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz}_{=1} = \sqrt{\pi}.
 \end{aligned}$$



Definition 2.

A random variable is said to have a **gamma distribution** with parameters (α, β) , $\alpha > 0, \beta > 0$, if its density function is given by

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\Gamma(\alpha) \beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}} & \text{if } x > 0 \\ 0 & \text{otherwise,} \end{cases}$$

where $\Gamma(\alpha)$, called the gamma function, is defined as

$$\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} e^{-t} dt.$$

We write $X \sim \text{Gamma}(\alpha, \beta)$.

Proposition 3.

Let $X \sim \text{Gamma}(\alpha, \beta)$, then $f(x; \alpha, \beta)$ is a well defined density, i.e.

$$\int_0^{\infty} \frac{1}{\Gamma(\alpha) \beta^{\alpha}} x^{\alpha-1} e^{-\frac{x}{\beta}} dx = 1.$$

Proof.

$$\begin{aligned} \int_{-\infty}^{\infty} f(x; \alpha, \beta) dx &= \int_0^{\infty} \frac{1}{\Gamma(\alpha) \beta^{\alpha}} x^{\alpha-1} e^{-\frac{x}{\beta}} dx \\ &= \int_0^{\infty} \frac{1}{\Gamma(\alpha)} w^{\alpha-1} e^{-w} dw \quad \text{by putting } w = \frac{x}{\beta} \\ &= \frac{1}{\Gamma(\alpha)} \int_0^{\infty} w^{\alpha-1} e^{-w} dw = \frac{1}{\Gamma(\alpha)} \times \Gamma(\alpha) = 1 \end{aligned}$$



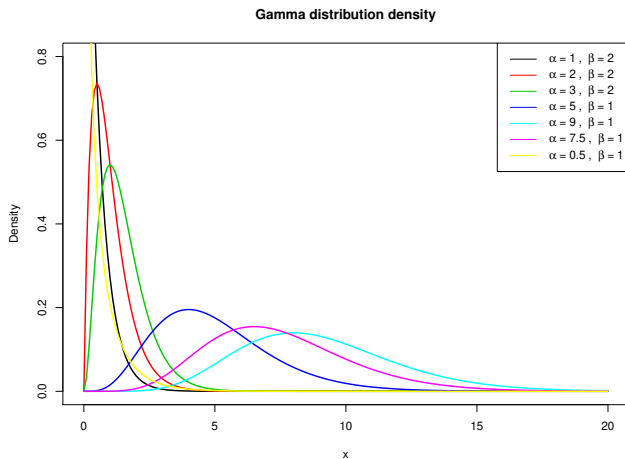


Figure 1: Density of Gamma distribution for various parameters

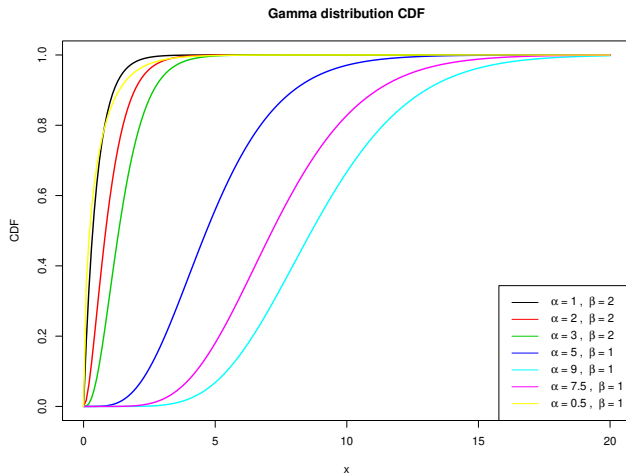


Figure 2: CDF of Gamma distribution for various parameters

Proposition 4.

If $X \sim \text{Gamma}(\alpha, \beta)$, then

$$\mathbb{E}[X] = \alpha\beta \quad \text{and} \quad \text{Var}(X) = \alpha\beta^2.$$

Proof.

$$\begin{aligned}\mathbb{E}(X) &= \int_0^\infty x \times \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}} dx \\&= \frac{1}{\Gamma(\alpha)} \int_0^\infty \frac{1}{\beta^\alpha} x^\alpha e^{-\frac{x}{\beta}} dx \\&= \frac{1}{\Gamma(\alpha)} \times \beta \int_0^\infty w^\alpha e^{-w} dw \quad \text{by putting } w = \frac{x}{\beta} \\&= \frac{1}{\Gamma(\alpha)} \times \beta \times \Gamma(\alpha + 1) = \frac{1}{\Gamma(\alpha)} \times \beta \times \alpha\Gamma(\alpha) = \alpha\beta\end{aligned}$$



Proof: Cont'd.

$$\begin{aligned}
\mathbb{E}[X^2] &= \int_0^\infty x^2 \times \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}} dx \\
&= \frac{1}{\Gamma(\alpha)} \int_0^\infty \frac{1}{\beta^\alpha} x^{\alpha+1} e^{-\frac{x}{\beta}} dx \\
&= \frac{1}{\Gamma(\alpha)} \times \beta^2 \int_0^\infty w^{\alpha+1} e^{-w} dw \quad \text{by putting } w = \frac{x}{\beta} \\
&= \frac{1}{\Gamma(\alpha)} \times \beta^2 \times \Gamma(\alpha+2) = \frac{1}{\Gamma(\alpha)} \times \beta \times (\alpha+1)\Gamma(\alpha+1) \\
&= \frac{1}{\Gamma(\alpha)} \times \beta^2 \times (\alpha+1)\alpha\Gamma(\alpha) = (\alpha+1)\alpha\beta^2
\end{aligned}$$

Hence,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = (\alpha+1)\alpha\beta^2 - (\alpha\beta)^2 = \alpha\beta^2.$$



Exercise 1.

Let X has Gamma distribution with parameters $\alpha = 2$ and $\beta = 3$;

1. Find the the probability that X exceeds 4;
2. find the mean and variance.

Solution 1.

1. $X \sim \text{Gamma}(2, 3)$, thus

$$\begin{aligned}
 P(X > 4) &= \int_4^{\infty} \frac{1}{\Gamma(2)3^2} x^{2-1} e^{-\frac{x}{3}} dx \\
 &= \frac{1}{9} \int_4^{\infty} x e^{-\frac{x}{3}} dx \\
 &= \left[-\frac{x}{3} e^{-\frac{x}{3}} \right] \Big|_4^{\infty} - \frac{1}{9} \int_4^{\infty} e^{-\frac{x}{3}} (-3) dx \\
 &= \frac{4}{3} e^{-\frac{4}{3}} - e^{-\frac{x}{3}} \Big|_4^{\infty} \\
 &= \frac{7}{3} e^{-\frac{4}{3}} = 0.615.
 \end{aligned}$$

2. $X \sim \text{Gamma}(2, 3)$, thus

$$\mu = \alpha\beta = 2 \cdot 3 = 6, \quad \sigma^2 = \alpha\beta^2 = 2 \cdot 3^2 = 18.$$

Exercise 2.

If a company employs n salespersons, its gross sales in thousands of dollars may be regarded as a random variable having a gamma distribution with $\alpha = 80\sqrt{n}$ and $\beta = 2$. If the sales cost is \$8,000 per salesperson, how many salespersons should the company employ to maximize the expected profit?

Solution 2.

When $X \sim \text{Gamma}(\alpha = 80\sqrt{n}, \beta = 2)$, then, the mean is given by

$$\mu = \alpha\beta = 80 \cdot 2\sqrt{n} = 160\sqrt{n}$$

Thus, the profit is

$$E = 160\sqrt{n} - 8n \implies$$

It maximum with respect to n , is then given by

$$\frac{dE}{dn} = \frac{160}{2\sqrt{n}} - 8 \implies n = 100$$

Proposition 5.

If $X \sim \text{Gamma}(\alpha, \beta)$, then

$$M_X(t) = \frac{1}{(1 - \beta t)^\alpha} \quad \text{for } t < \frac{1}{\beta}.$$

Proof.

$$\begin{aligned}
 M_X(t) &= \mathbb{E} \left[e^{tX} \right] = \int_0^\infty e^{tx} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}} dx \\
 &= \int_0^\infty \frac{1}{\Gamma(\alpha)\beta^\alpha} e^{-x\left(\frac{1}{\beta}-t\right)} x^{\alpha-1} dx \\
 \text{put } y &= x \left(\frac{1}{\beta} - t \right) = \int_0^\infty \frac{1}{\Gamma(\alpha)\beta^\alpha} y^{\alpha-1} e^{-y} \left(\frac{\beta}{1-\beta t} \right)^{\alpha-1} \left(\frac{\beta}{1-\beta t} \right) dy \\
 &= \frac{1}{\Gamma(\alpha)\beta^\alpha} \left(\frac{\beta}{1-\beta t} \right)^\alpha \int_0^\infty y^{\alpha-1} e^{-y} dy \\
 &= \frac{1}{\Gamma(\alpha)\beta^\alpha} \left(\frac{\beta}{1-\beta t} \right)^\alpha \Gamma(\alpha) \\
 &= \frac{1}{(1-\beta t)^\alpha}.
 \end{aligned}$$



Definition 4 (Exponential Random Variable).

A random variable X has an **exponential distribution** and it is referred to as an exponential random variable if and only if its probability density is given by

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & ; \quad x > 0 \\ 0 & ; \quad \text{elsewhere} \end{cases}$$

It is denoted as

$$X \sim \text{Exp}(\lambda)$$

Alternatively, X can be viewed as a particular case of the Gamma distribution with $\alpha = 1$, and density

$$f(x; \beta) = \begin{cases} \frac{1}{\beta} e^{-\frac{x}{\beta}} & ; \quad x > 0 \\ 0 & ; \quad \text{elsewhere} \end{cases}$$

Theorem 5.

The cumulative distribution function of a exponential random variable X with parameter λ is

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & ; \quad x > 0 \\ 0 & ; \quad x \leq 0 \end{cases}$$

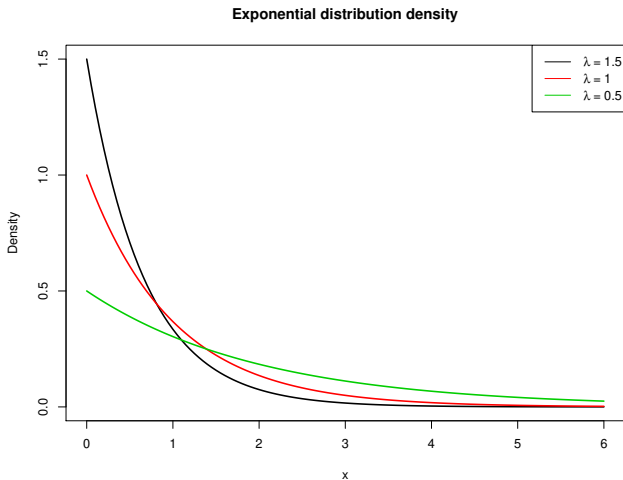


Figure 3: Density of exponential distribution for various value of λ

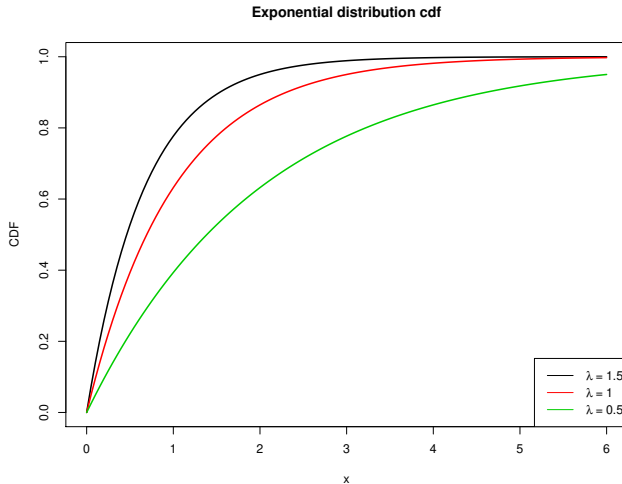


Figure 4: CDF of exponential distribution for various value of λ

Proposition 6 (Properties).

For an exponential random variable $X \sim \text{Exp}(\lambda)$.

- *Moment generating function:*

$$M_X(t) = \frac{\lambda}{\lambda - t} \quad \text{for } t < \lambda$$

- *Expectation and variance are:*

$$\mathbb{E}[X] = \frac{1}{\lambda} \quad \text{and} \quad \mathbb{V}\text{ar}(X) = \frac{1}{\lambda^2}.$$

Proof.

- For the mgf,

$$\mathbb{E} \left[e^{tX} \right] = \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx = \frac{\lambda}{\lambda - t} \underbrace{\int_0^{\infty} (\lambda - t) e^{-(\lambda - t)x} dx}_{= 1} = \frac{\lambda}{\lambda - t},$$

for $t < \lambda$,

- For the expectation, we have

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \lambda e^{-\lambda x} dx \\ &\stackrel{\text{IBP}}{=} \left[-x e^{-\lambda x} \right]_{x=0}^{\infty} + \int_0^{\infty} e^{-\lambda x} dx \\ &= (0 - 0) + \frac{1}{\lambda} \underbrace{\int_0^{\infty} \lambda e^{-\lambda x} dx}_{= 1} = \frac{1}{\lambda} \end{aligned}$$



Proof: Cont'd.

For the variance

$$\begin{aligned}\mathbb{E}[X^2] &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx \\ &\stackrel{\text{IBP}}{=} \left[-x^2 e^{-\lambda x} \right]_{x=0}^{\infty} + \int_0^{\infty} 2x e^{-\lambda x} dx \\ &= (0 - 0) + \underbrace{\frac{2}{\lambda} \int_0^{\infty} x \lambda e^{-\lambda x} dx}_{= \mathbb{E}[X] = \frac{1}{\lambda}} = \frac{2}{\lambda^2}.\end{aligned}$$

we conclude

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$



Definition 3.

Let X follow a gamma distribution with $\alpha = \frac{n}{2}$ and $\beta = 2$ (i.e. $X \sim \text{Gamma}(\frac{n}{2}, 2)$), where n is a positive integer. Then the probability density function of X is:

$$f(x) = \begin{cases} \frac{1}{\Gamma(\frac{n}{2})2^{\frac{n}{2}}} x^{\frac{n}{2}-1} e^{-x/2} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

We say that X follows a **chi-square distribution** with n degrees of freedom, denoted $X \sim \chi_n^2$ and read *chi-square- n* .

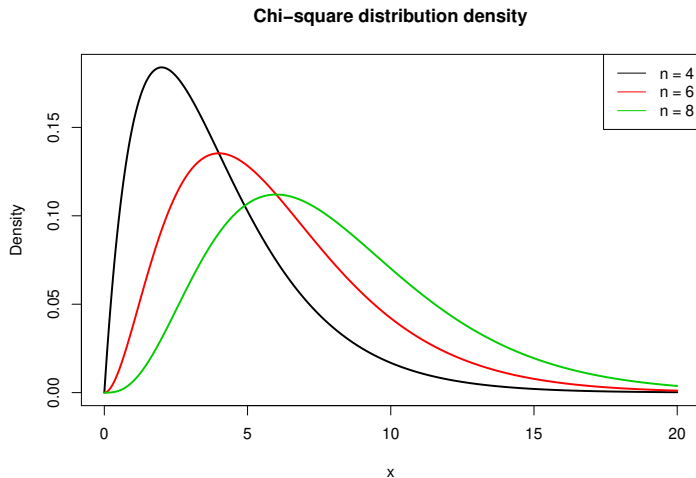


Figure 5: Density of Chi-square distribution for various degrees of freedom.

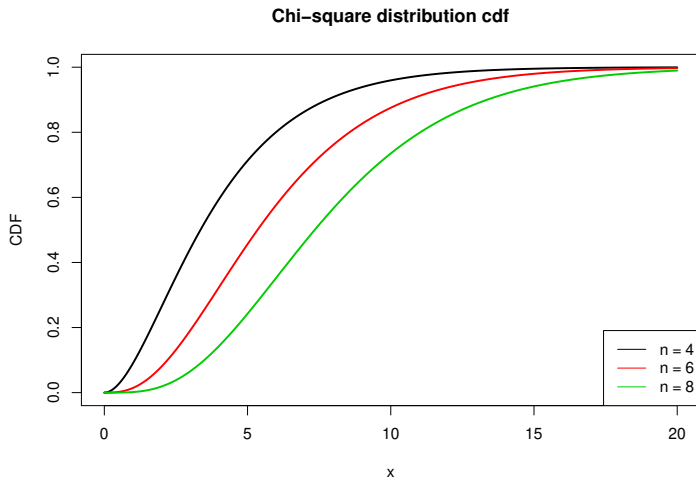


Figure 6: CDF of Chi-square distribution for various degrees of freedom.

Corollary 1.

If $X \sim \chi_n^2$, then,

1. $\mathbb{E}[X] = n$;
2. $\text{Var}(X) = 2n$;
3. *Mgf*

$$M_X(t) = \frac{1}{(1 - 2t)^{n/2}}.$$

Definition 4.

A random variable is said to have a beta distribution if its density is given by

$$f(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise,} \end{cases}$$

where $\alpha, \beta > 0$, and

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx,$$

is the Beta function. We write $X \sim \text{Beta}(\alpha, \beta)$ to indicate such distribution.

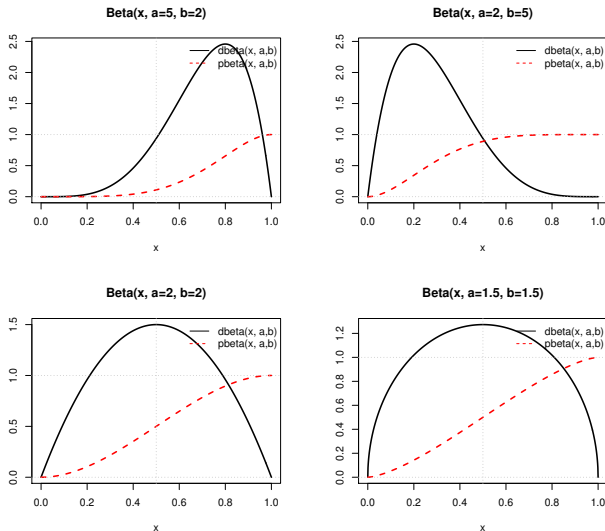


Figure 7: CDF and PDF of beta distribution for various values of parameters.

Theorem 2.

We have,

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$

Proposition 7.

If $X \sim \text{Beta}(\alpha, \beta)$, then

1.

$$\mathbb{E}[X^k] = \frac{B(\alpha + k, \beta)}{B(\alpha, \beta)}, \quad k > -\alpha.$$

2.

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

3.

$$M_X(t) = 1 + \sum_{k=1}^{\infty} \left(\prod_{r=0}^{k-1} \frac{\alpha + r}{\alpha + \beta + r} \right) \frac{t^k}{k!}.$$

Proof.

1. For the k -th moment, we have

$$\begin{aligned}
 \mathbb{E}[X^k] &= \int_0^1 x^k f_X(x) \, dx = \frac{1}{B(\alpha, \beta)} \int_0^1 x^k x^{\alpha-1} (1-x)^{\beta-1} \, dx \\
 &= \frac{1}{B(\alpha, \beta)} \underbrace{\int_0^1 x^{\alpha+k-1} (1-x)^{\beta-1} \, dx}_{=B(\alpha+k, \beta)} \\
 &= \frac{B(\alpha+k, \beta)}{B(\alpha, \beta)}
 \end{aligned}$$

2. By taking $k = 1$ in (1), we get

$$\begin{aligned}
 \mathbb{E}[X] &= \frac{B(\alpha+1, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \\
 &= \frac{\alpha\Gamma(\alpha)\Gamma(\beta)}{(\alpha+\beta)\Gamma(\alpha+\beta)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} = \frac{\alpha}{\alpha+\beta}.
 \end{aligned}$$



Proof: Cont'd.

2. By taking $k = 2$ in (1), we get

$$\begin{aligned}
 \mathbb{E}[X^2] &= \frac{B(\alpha+2, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+\beta+2)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \\
 &= \frac{\alpha(\alpha+1)\Gamma(\alpha)\Gamma(\beta)}{(\alpha+\beta)(\alpha+\beta+1)\Gamma(\alpha+\beta)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \\
 &= \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}.
 \end{aligned}$$

Thus,

$$\begin{aligned}
 \text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} - \left(\frac{\alpha}{\alpha+\beta}\right)^2 \\
 &= \frac{\alpha(\alpha^2\alpha\beta + \alpha + \beta - \alpha^2 - \alpha\beta - \alpha)}{(\alpha+\beta)^2(\alpha+\beta+1)} \\
 &= \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.
 \end{aligned}$$



Proof: Cont'd.

3.

$$\begin{aligned}
M_X(t) &= \int_0^1 e^{tx} f_X(x) dx = \frac{1}{B(\alpha, \beta)} \int_0^1 e^{tx} x^{\alpha-1} (1-x)^{\beta-1} dx \\
&= \frac{1}{B(\alpha, \beta)} \int_0^1 \left(\sum_{k=0}^{\infty} \frac{(tx)^k}{k!} \right) x^{\alpha-1} (1-x)^{\beta-1} dx \\
&= \frac{1}{B(\alpha, \beta)} \sum_{k=0}^{\infty} \frac{t^k}{k!} \int_0^1 x^{\alpha+k-1} (1-x)^{\beta-1} dx \\
&= \sum_{k=0}^{\infty} \frac{t^k}{k!} \left(\frac{B(\alpha+k, \beta)}{B(\alpha, \beta)} \right) \\
&= \frac{B(\alpha, \beta)}{B(\alpha, \beta)} \frac{t^0}{0!} + \sum_{k=1}^{\infty} \frac{t^k}{k!} \left(\frac{B(\alpha+k, \beta)}{B(\alpha, \beta)} \right) \\
&= 1 + \sum_{k=1}^{\infty} \left(\frac{\Gamma(\alpha+k) \Gamma(\beta)}{\Gamma(\alpha+\beta+k)} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha) \Gamma(\beta)} \right) \frac{t^k}{k!}
\end{aligned}$$



Proof: Cont'd.

3.

$$\begin{aligned}
 M_X(t) &= 1 + \sum_{k=1}^{\infty} \left(\frac{\Gamma(\alpha + k)}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha + \beta + k)} \right) \frac{t^k}{k!} \\
 &= 1 + \sum_{k=1}^{\infty} \left(\frac{\Gamma(\alpha) \prod_{r=0}^k (\alpha + r)}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha + \beta) \prod_{r=0}^k (\alpha + \beta + r)} \right) \frac{t^k}{k!} \\
 &= 1 + \sum_{k=1}^{\infty} \left(\prod_{r=0}^{k-1} \frac{\alpha + r}{\alpha + \beta + r} \right) \frac{t^k}{k!}.
 \end{aligned}$$



Exercise 3.

If the annual proportion of new restaurants that fail in a given city may be looked upon as a random variable having a beta distribution with $\alpha = 1$ and $\beta = 4$, find

- (a) the mean of this distribution, that is, the annual proportion of new restaurants that can be expected to fail in the given city;
- (b) the probability that at least 25 percent of all new restaurants will fail in the given city in any one year.

Solution 3.

Given that $\alpha = 1$ and $\beta = 4$

(a) the expected value is

$$\mu = \frac{1}{1+4} = \frac{1}{5}$$

(b) the requested probability is

$$\begin{aligned} \mathbb{P}(X \geq 0.25) &= \frac{\Gamma(5)}{\Gamma(1)\Gamma(4)} \int_{0.25}^1 (1-x)^3 dx \stackrel{y=1-x}{=} 4 \int_0^{0.75} y^3 dy \\ &= 4 \cdot \frac{y^4}{4} \Big|_0^{0.75} = (0.75)^4 = 0.3164 \end{aligned}$$

- The normal distribution is by far the most important probability distribution.
- One of the main reasons for that is the Central Limit Theorem (CLT).
- To give you an idea, the CLT states that if you add a large number of random variables, the distribution of the sum will be approximately normal under certain conditions.
- The importance of this result comes from the fact that many random variables in real life can be expressed as the sum of a large number of random variables and, by the CLT, we can argue that distribution of the sum should be normal.
- The CLT is one of the most important results in probability. Here, we will introduce normal random variables

Definition 5.

The continuous random variable X follows a **normal distribution** with parameters μ and σ if its probability density function is defined as:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right\}$$

for $-\infty < x < \infty$, $-\infty < \mu < \infty$ and $0 < \sigma < \infty$. we denote $X \sim \mathcal{N}(\mu, \sigma^2)$.

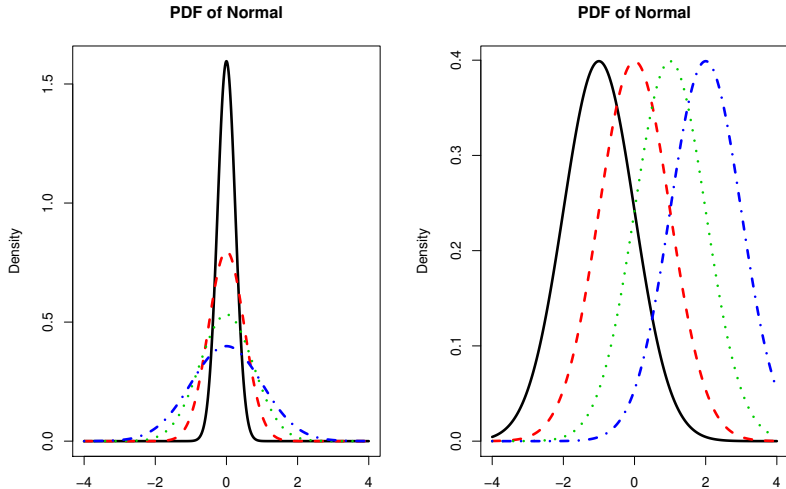
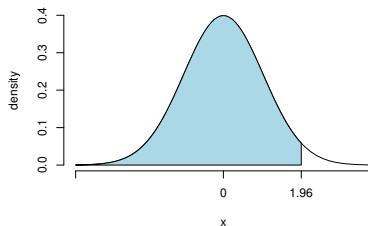


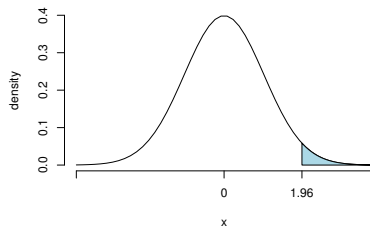
Figure 8: Effect of μ and σ on the standard normal pdf curve

The normal distribution

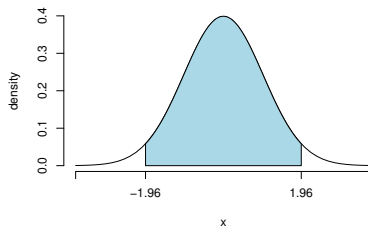
Normal Curve, mean = 0 , SD = 1
Shaded Area = 0.975



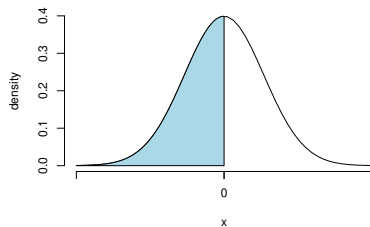
Normal Curve, mean = 0 , SD = 1
Shaded Area = 0.025



Normal Curve, mean = 0 , SD = 1
Shaded Area = 0.95



Normal Curve, mean = 0 , SD = 1
Shaded Area = 0.5



The normal distribution

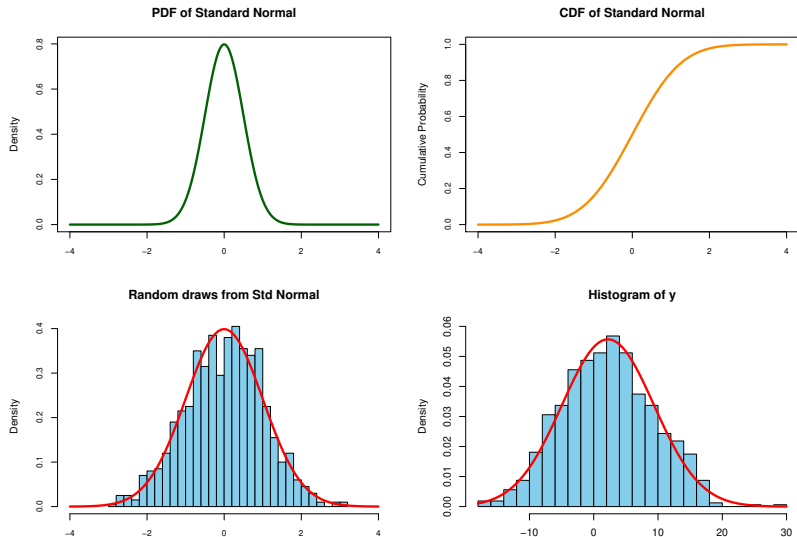


Figure 9: Illustration of the normal pdf and cdf

Proposition 8.

Let $X \sim \mathcal{N}(\mu, \sigma^2)$, then f_X is a well defined pdf, i.e.

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx = 1.$$

Proof: Let $X, Y \sim \mathcal{N}(\mu, \sigma^2)$ two normal random variables, then

$$\begin{aligned}
 \left(\int_{-\infty}^{\infty} f_X(x) dx \right) \left(\int_{-\infty}^{\infty} f_Y(y) dy \right) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma^2} e^{\left(-\frac{(x-\mu)^2 + (y-\mu)^2}{2\sigma^2} \right)} dx dy \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2 + v^2}{2} \right) du dv \quad \left(\text{where, } u = \frac{x-\mu}{\sigma}, v = \frac{y-\mu}{\sigma} \right) \\
 &= \frac{1}{2\pi} \int_0^{2\pi} \int_0^{\infty} \exp\left(-\frac{r^2}{2} \right) r dr d\theta \quad (\text{polar coordinates}) \\
 &= \int_0^{\infty} r \exp\left(-\frac{r^2}{2} \right) dr \\
 &= \left[-\exp\left(-\frac{r^2}{2} \right) \right]_{r=0}^{\infty} \\
 &= 1.
 \end{aligned}$$

Since $\int_{-\infty}^{\infty} f_X(x) dx = \int_{-\infty}^{\infty} f_Y(y) dy \geq 0$, the conclusion follows.

Proposition 9.

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$.

Proof.

$$\begin{aligned}\mathbb{E}[X] &= \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \\ &\stackrel{z=\frac{x-\mu}{\sigma}}{=} \int_{-\infty}^{\infty} (\sigma z + \mu) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz \\ &= \sigma \underbrace{\int_{-\infty}^{\infty} \frac{z}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz}_{=0} + \mu \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz}_{=1} \\ &= \mu.\end{aligned}$$



Proof.

$$\begin{aligned}
 \text{Var}(X) &= \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx \\
 &\stackrel{z = \frac{x - \mu}{\sigma}}{=} \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 \exp\left(-\frac{z^2}{2}\right) dz \\
 &\stackrel{\text{symmetry}}{=} \sigma^2 \sqrt{\frac{2}{\pi}} \int_0^{\infty} z^2 \exp\left(-\frac{z^2}{2}\right) dz \\
 &\stackrel{u = z^2/2}{=} \sigma^2 \frac{2}{\sqrt{\pi}} \underbrace{\int_{-\infty}^{\infty} u^{1/2} e^{-u} du}_{= \Gamma(\frac{3}{2}) = \frac{1}{2} \Gamma(\frac{1}{2}) = \frac{\sqrt{\pi}}{2}} = \sigma^2.
 \end{aligned}$$



Proposition 10.

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$Y = aX + b \sim \mathcal{N}(a\mu + b, a^2 \sigma^2).$$

In particular, if $Z = \frac{X - \mu}{\sigma}$, then $Z \sim \mathcal{N}(0, 1)$.

Proof.

$$\begin{aligned}
 P(Y \leq y) &= P(aX + b \leq y) = P\left(X \leq \frac{y-b}{a}\right) \\
 &= \int_{-\infty}^{\frac{y-b}{a}} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \\
 &\stackrel{z=\frac{ax+b}{a}}{=} \int_{-\infty}^y \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\left(\frac{z-b}{a}-\mu\right)^2}{2\sigma^2}\right) \frac{1}{a} dz \\
 &= \int_{-\infty}^y \underbrace{\frac{1}{\sqrt{2\pi a^2 \sigma^2}} \exp\left(-\frac{(z-(a\mu+b))^2}{2a^2 \sigma^2}\right)}_{=\text{pdf of } \mathcal{N}(a\mu+b, a^2 \sigma^2)} dz
 \end{aligned}$$

□

Proposition 11.

- If $Z \sim \mathcal{N}(0, 1)$, then

$$M_Z(t) = \exp\left(\frac{t^2}{2}\right).$$

- If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

Proof.

$$\begin{aligned}
 \mathbb{E} \left[e^{tZ} \right] &= \int_{-\infty}^{\infty} e^{tz} \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{z^2}{2}\right)} dz \\
 &= e^{t^2/2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{z^2}{2} + \frac{2tz}{2} - \frac{t^2}{2} \right) dz \\
 &= \exp \left(\frac{t^2}{2} \right) \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{(z-t)^2}{2} \right) dz}_{=1} = \exp \left(\frac{t^2}{2} \right).
 \end{aligned}$$

for the second, note that $X \stackrel{\text{law}}{=} \mu + \sigma Z$, then

$$M_X(t) = \mathbb{E} \left[e^{t(\mu + \sigma Z)} \right] = e^{t\mu} \mathbb{E} \left[e^{(t\sigma)Z} \right] = \exp \left(t\mu + \frac{(t\sigma)^2}{2} \right).$$



Proposition 12.

If X is normally distributed with mean μ and variance $\sigma^2 > 0$, then:

$$U = \left(\frac{X - \mu}{\sigma} \right)^2 = Z^2$$

is distributed as a chi-square random variable with 1 degree of freedom (i.e. χ_1^2).

Proof.

We will use the cdf method

$$\begin{aligned}F_U(u) &= \mathbb{P}(U \leq u) = \mathbb{P}(Z^2 \leq u) = \mathbb{P}(-\sqrt{u} < Z < \sqrt{u}) \\&= \int_{-\sqrt{u}}^{\sqrt{u}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz = 2 \int_0^{\sqrt{u}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz \\&= 2 \int_0^u \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y}{2}\right) \left(\frac{1}{2\sqrt{y}}\right) dy, \quad \text{By putting } y = z^2 \\&= \int_0^u \frac{1}{\sqrt{\pi}\sqrt{2}} y^{\frac{1}{2}-1} \exp\left(-\frac{y}{2}\right) dy \\&= \int_0^u \frac{1}{\Gamma(1/2)2^{1/2}} y^{\frac{1}{2}-1} \exp\left(-\frac{y}{2}\right) dy = \int_0^u f_U(y) dy,\end{aligned}$$

where f_U is the pdf of χ_1^2 .



Theorem 3.

If X is a random variable having a binomial distribution with the parameters n and θ , then

$$Z = \frac{X - n\theta}{\sqrt{n\theta(1 - \theta)}}$$

approaches that of the standard normal distribution when $n \rightarrow \infty$.

Example 6.

Let X be the number of times that a fair coin that is flipped 40 times lands on heads. Find the probability that $X = 20$. Use the normal approximation and then compare it with the exact solution.

Solution

- To employ the normal approximation, note that because the binomial is a discrete integer-valued random variable, whereas the normal is a continuous random variable, it is best to write $P(X = k)$ as $P(k - 1/2 < X < k + 1/2)$ before applying the normal approximation (this is called the continuity correction). Doing so gives

$$\begin{aligned}P(X = 20) &= P(19.5 \leq X < 20.5) = P\left(\frac{19.5 - 20}{\sqrt{10}} < \frac{X - 20}{\sqrt{10}} < \frac{20.5 - 20}{\sqrt{10}}\right) \\&\approx P\left(-0.16 < \frac{X - 20}{\sqrt{10}} < 0.16\right) \\&\approx \Phi(0.16) - \Phi(-0.16) \approx 0.1272\end{aligned}$$

- The exact result is

$$P(X = 20) = \binom{40}{20} \left(\frac{1}{2}\right)^{40} \approx 0.1254$$

Exercise 4.

A randomly chosen IQ test taker obtains a score that is approximately a normal random variable with mean 100 and standard deviation 15. Find the probability that the score of such a person is

1. above 125.
2. between 90 and 110.

Solution 4.

1. Let X be the obtained score and let

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 100}{15} \implies Z \sim \mathcal{N}(0, 1)$$

Therefore,

$$\mathbb{P}(X > 125) = \mathbb{P}\left(Z > \frac{125 - 100}{15}\right) = \mathbb{P}\left(Z > \frac{5}{3}\right) = 1 - \Phi\left(\frac{5}{3}\right) \approx 0.048$$

2. The requested probability is

$$\begin{aligned}\mathbb{P}(90 < X < 100) &= \mathbb{P}\left(\frac{90 - 100}{15} < Z < \frac{110 - 100}{15}\right) \\ &= \mathbb{P}\left(\frac{-2}{3} < Z < \frac{2}{3}\right) = \Phi\left(\frac{2}{3}\right) - \Phi\left(-\frac{2}{3}\right) \\ &= \Phi\left(\frac{2}{3}\right) - \left(1 - \Phi\left(\frac{2}{3}\right)\right) = 2\Phi\left(\frac{2}{3}\right) - 1 \approx 0.495\end{aligned}$$

Exercise 5.

The life of a certain type of automobile tire is normally distributed with mean 34,000 miles and standard deviation 4,000 miles.

1. What is the probability that such a tire lasts over 40,000 miles?
2. What is the probability that such a tire lasts between 30,000 and 35,000 miles?
3. Given that such a tire has survived 30,000 miles, what is the conditional probability that the tire survives another 10,000 miles?

Solution 5.

1. Let X be the life of the tire, then

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 34000}{4000} \sim \mathcal{N}(0, 1)$$

Therefore,

$$\begin{aligned}\mathbb{P}(X > 40000) &= \mathbb{P}\left(Z > \frac{40000 - 34000}{4000}\right) \\ &= \mathbb{P}\left(Z > \frac{3}{2}\right) = 1 - \Phi\left(\frac{3}{2}\right)\end{aligned}$$

2. The requested probability

$$\begin{aligned}\mathbb{P}(30000 < X < 35000) &= \mathbb{P}\left(\frac{30000 - 34000}{4000} < Z < \frac{35000 - 34000}{4000}\right) \\ &= \mathbb{P}\left(-1 < Z < \frac{1}{4}\right) = \Phi\left(\frac{1}{4}\right) - \Phi(-1) \approx 0.44\end{aligned}$$

Solution 5.

3. Given that

$$\begin{aligned}\mathbb{P}(X > 30000) &= \mathbb{P}\left(Z > \frac{30000 - 34000}{4000}\right) \\ &= \mathbb{P}(Z > -1) = 1 - \Phi(-1) \approx 0.841 \\ \mathbb{P}(X > 40000) &= \mathbb{P}\left(Z > \frac{40000 - 34000}{4000}\right) \\ &= \mathbb{P}\left(Z > \frac{3}{2}\right) = 1 - \Phi\left(\frac{3}{2}\right) \approx 0.067\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbb{P}(X > 40000 \mid X > 30000) &= \frac{\mathbb{P}(X > 40000 \cap X > 30000)}{\mathbb{P}(X > 30000)} \\ &= \frac{\mathbb{P}(X > 40000)}{\mathbb{P}(X > 30000)} \\ &= \frac{1 - \Phi\left(\frac{3}{2}\right)}{1 - \Phi(1)} = \frac{0.067}{0.841} \approx 0.0796\end{aligned}$$

Exercise 6.

The annual rainfall in Cleveland, Ohio, is approximately a normal random variable with mean 30.2 inches and standard deviation 8.4 inches. Let A_i be the event that the rainfall in the next i th year exceeds 44 inches, and assume that all A_i are independent. Find the probability that

1. next year's rainfall will exceed 44 inches?
2. the yearly rainfall in exactly 3 of the next 7 years will exceed 44 inches?

Solution 6.

1. Let X be the rainfall in a year. Then,

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 30.2}{8.4} \sim \mathcal{N}(0, 1)$$

Therefore,

$$\begin{aligned} \mathbb{P}(X > 44) &= \mathbb{P}\left(Z > \frac{44 - 30.2}{8.4}\right) = \mathbb{P}\left(Z > \frac{13.8}{8.4}\right) \\ &= \mathbb{P}\left(Z > \frac{23}{14}\right) = 1 - \Phi\left(\frac{23}{14}\right) \approx 0.050 \end{aligned}$$

2. For any year, as the rainfall in every year is independent of the previous years, the probability that the rainfall will exceed 44 inches is

$$\mathbb{P}(X > 44) = 1 - \Phi\left(\frac{23}{14}\right)$$

Therefore, the probability that the rainfall will exceed 44 inches in exactly 3 of the next 7 years is $\binom{7}{3} \left(1 - \Phi\left(\frac{23}{14}\right)\right)^3 \left(\Phi\left(\frac{23}{14}\right)\right)^4$.

Exercise 7.

A normal random variable X has an unknown mean μ and a standard deviation $\sigma = 2.5$. If the probability that X exceeds 7.5 is 0.8289, find μ .

Solution 7.

$$\begin{aligned}P(X > 7.5) = 0.8289 &\iff P\left(Z > \frac{7.5 - \mu}{2.5}\right) = 0.8289 \\&\iff P\left(Z \leq \frac{7.5 - \mu}{2.5}\right) = 1 - 0.8289 = 0.1711 \\&\iff \frac{7.5 - \mu}{2.5} = -0.95 \\&\iff 7.5 - \mu = (-0.95) \times (2.5) = -2.375 \\&\iff \mu = 7.5 + 2.375 = 9.875.\end{aligned}$$

Exercise 8.

A random variable has a normal distribution with $\sigma = 10$. If the probability that the random variable will take on a value less than 82.5 is 0.8212, what is the probability that it will take on a value greater than 58.3?

Solution 8.

Given that $X \sim \mathcal{N}(\mu, \sigma = 10)$, then

$$\begin{aligned}\mathbb{P}(X < 82.5) = 0.8212 &\iff \mathbb{P}\left(Z < \frac{82.5 - \mu}{10}\right) = 0.8212 \\ &\iff \frac{82.5 - \mu}{10} = 0.92 \\ &\iff 82.5 - \mu = (0.92) \times (10) = 9.2 \\ &\iff \mu = 82.5 - 9.2 = 73.3.\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbb{P}(X > 58.3) &= \mathbb{P}\left(Z > \frac{58.3 - 73.3}{10}\right) = \mathbb{P}(Z > -1.5) \\ &= 1 - P(Z \leq -1.5) = 0.9332\end{aligned}$$

Standard normal distribution table

	Standard Normal Distribution Table									
	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Standard normal distribution table

	Standard Normal Distribution Table									
	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-1	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-2	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-3	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010

Main textbook:

- John E. Freund's. *Mathematical Statistics With Applications* Ed: 8 (Required).

Suggested textbooks:

- Ross, S., 2009. *A first course in probability*. Upper Saddle River, 6.
- Hossein Pishro-Nik. *Probability, Statistics and Random Processes* [here](#).

Some useful websites:

- [Society of Actuaries, Exam of probability, sample questions](#).
- [Society of Actuaries, Exam of probability, sample solutions](#).